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1- DESCRIPTION

This procedure describes the removal and replacement of the Twin Resonator Module (TRM) in systems **with** LCC, and WideOpen Enclosure Magnets. The TRM uses the BRM-D parts for installation/removal.

2- LIST OF EQUIPMENT & PARTS

Note

There are 2 revisions of Gradient Insertion tools. The original kit (2164744) is used to replace a BRM coil. The upgraded kit (2164744-3) can be used to replace BRM, coils, CRM coils and BRM-D coils.

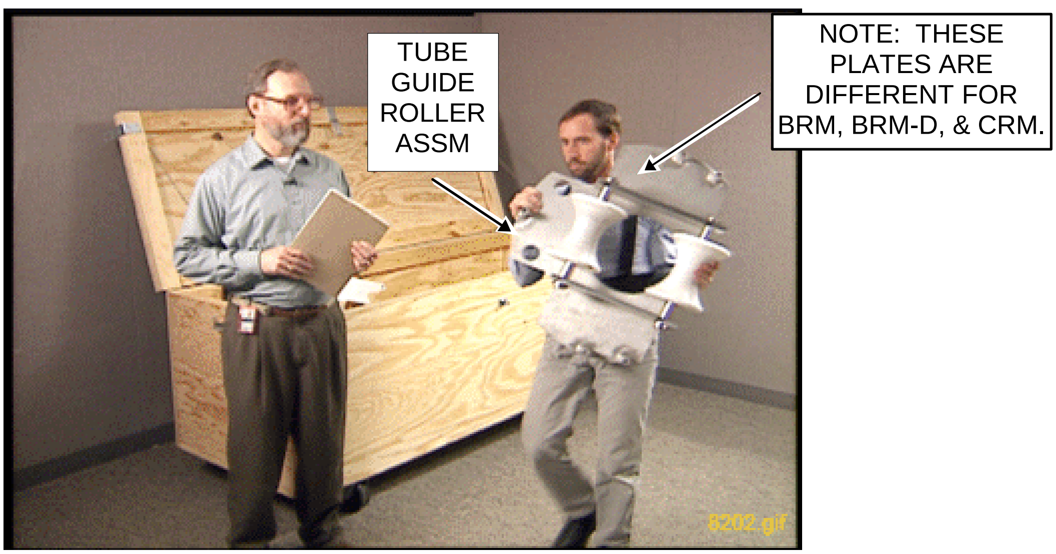
Item #	Description	Quantity	Part Number
1.	Gradient Insertion Tool Kit	1	2164744-3
	<u>Kit contents:</u>		
	Tube Support Plate Assembly	1	2284928
	Male Support Tube Assembly (weighs 80 lbs)	1	2284929
	Female Support Tube Assembly (weighs 60lbs)	1	2284930
	Tube Guide Roller Assembly	2	2164707
	Tube Jacking Assembly	1	2290367
	Crate	1	2172196
	Hex Head cap screws M10 x 80 stainless	10	46-318508P59
	Hex Head cap screws M10 x 25 stainless	4	46-318508P20
	M10 x 120 stainless steel studs	16	2180498
	M10 stainless steel nuts	24	2109875-4
	M10 stainless steel flat washers	24	2109878-4
	Hex Head Cap Screw M10 x 50 stainless	10	46-318508P25
	Washer Plain – Normal 13 mm 24 mm	8	46-328430P8
	Hairpin Cotter, 1/8 Dia Wire, stainless	2	46-252065P118
	Stud stainless, M10 x 65	6	2116325-2
	BRM Roller/Bracket Assemblies	2	2121111
	CRM Roller/Bracket Assemblies	2	2187590
	BRM-D Roller/Bracket Assemblies (Pat. End)	1	2213270
	BRM-D Roller/Bracket Assemblies (Svc. End)	1	2213270-2
	Bolts for BRM Rollers	10	2109866-24
	3/8 x ¾ x 3” Plain Steel Spacer	2	2224996
	3/8 – 16 Steel Nut Wing	8	2224996
	Rope (10 ft long with snap clips)	1	2188986
	BRM Insertion Tool Ident Label	1	2280510
2.	BRM/BRM-D/CRM Cart	1	2134810
3.	Aluminum Cradle	1	2134810-2
4.	Cable Crimper/Stripper Kit	1	2134776
5.	Poron Seal, for air cover	1	2185175
6.	Poron Seal,	4	2181231
7.	Poron Seal	8	2181231-2
8.	Red Loctite # 271	1	46-170686p3
9.	Blue Loctite # 242	1	46-170684p2
10.	Alcohol	1	Field Supplied
11.	Ty-wraps	5	Field Supplied
12.	Splice kit	1	2241521

List of Non-Magnetic Tools

1. 3/8 inch T-bar allen wrench
2. 5/16 inch right angle allen wrench
3. 8 mm allen wrench
4. 8 inch adjustable wrench
5. 17 mm open end wrench
6. 10 mm socket
7. 13 mm socket
8. 17 mm socket
9. 19 mm socket
10. 24 mm socket
11. 3/8 inch drive ratchet
12. 1/2 inch drive ratchet
13. 3 inch long extension for 1/2 inch drive
14. 3 Foot Long Level
15. Phillips Head Screwdriver
16. Standard Flat Head Screwdriver

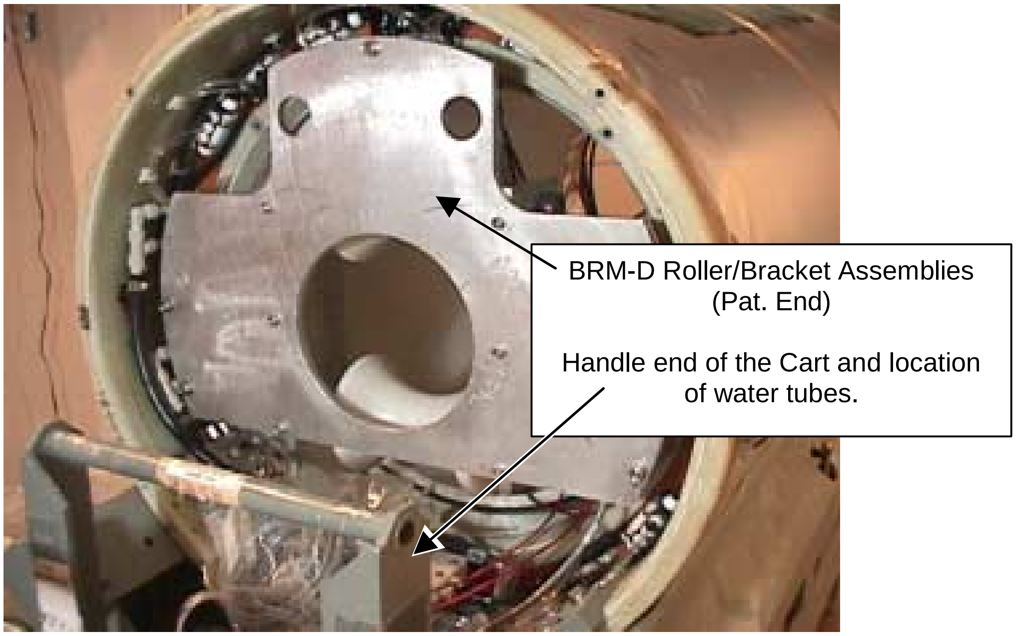
3- INSTALLING THE TRM

1. Remove the two Tube Guide Roller Assemblies from the crate. You may need to exchange the rollers to the BRM-D Plate Assembly. Install one Assembled Tube Guide Roller Assembly onto each end of the Gradient Coil. See Illustration 3-1. The kit contains 16 M10 x 120 studs. Use these studs at the top hole location. The stud will support the Tube Guide Roller Assembly and will assist the alignment of the remaining bolts.



TUBE GUIDE ROLLER ASSEMBLY
ILLUSTRATION 3-1

2. Install the Patient End plate (p/n 2213270) to the coil. See Illustration 3-2.



PATIENT END PLATE ON COIL
ILLUSTRATION 3-2

3. Secure plate, using washer and nut on studs, see Illustration 3-3.

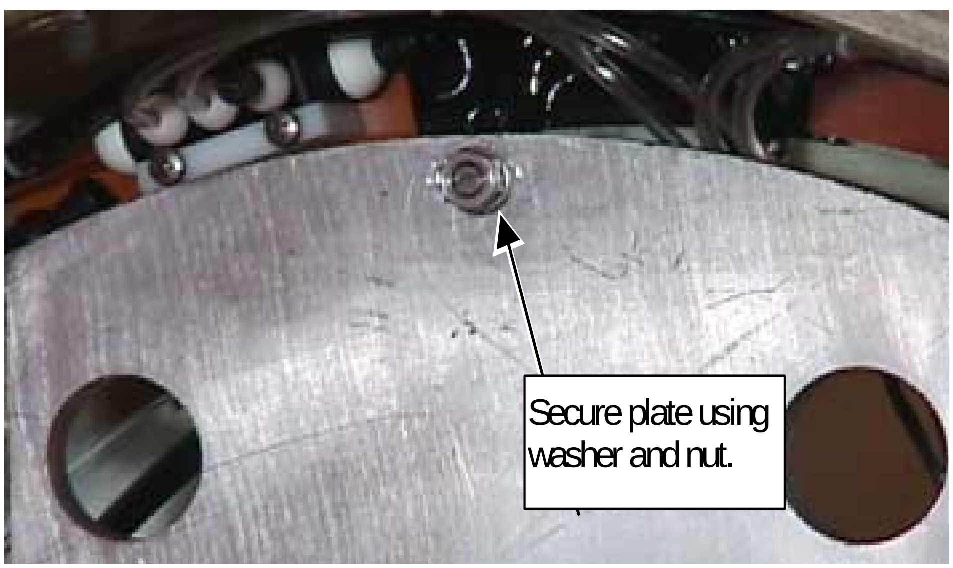
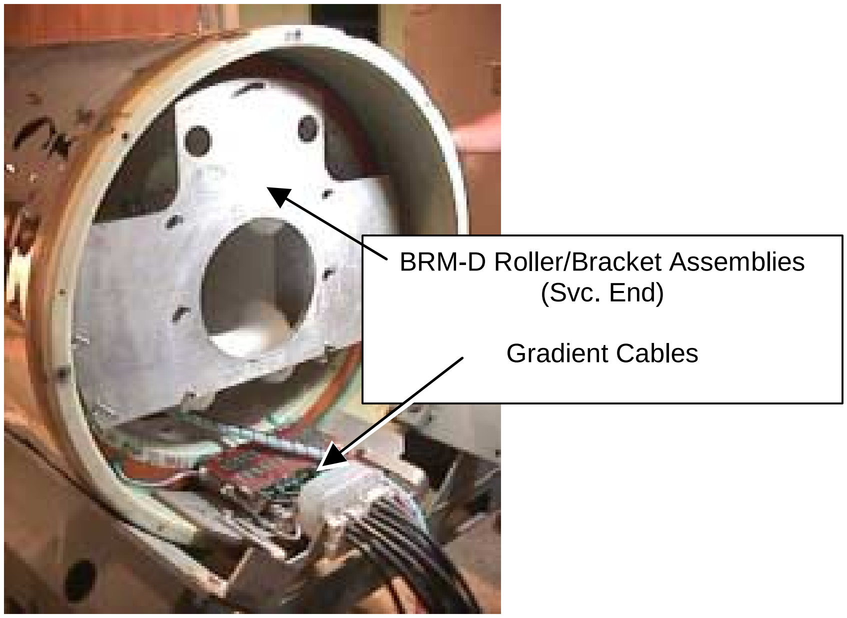


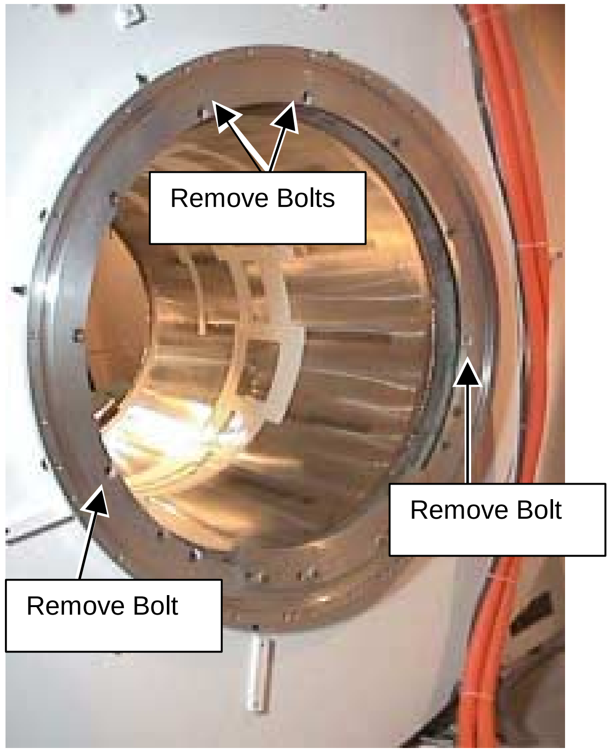
PLATE SECURED TO COIL
ILLUSTRATION 3-3

4. Install the Service End plate (p/n 2213270-2). See Illustration 3-4.



SERVICE END PLATE ON COIL
ILLUSTRATION 3-4

5. Go to the rear of the magnet and Install the Tube Support Plate onto the end flange.
   a. Remove screws from the interface ring so the Tube Support Plate can be installed. See Illustration 3-5.



LOCATION OF BOLTS TO BE REMOVED ON REAR OF MAGNET INTERFACE RING
ILLUSTRATION 3-5

b. Place the four of the M10 x 60 studs in the interface ring openings from step a, see Illustration 3-6.

PLACEMENT OF STUD ON THE INTERFACE RING
ILLUSTRATION 3-6

c. Place bushing on stud. See Illustration 3-7.

IMPORTANT

YOU MUST USE BUSHINGS, OR DAMAGE COULD OCCUR.

PLACING BUSHING ON STUD
ILLUSTRATION 3-7

d. Tighten stud to support the weight of the Tube Support Plate Assembly, see Illustration 3-8.

TIGHTENING STUD
ILLUSTRATION 3-8

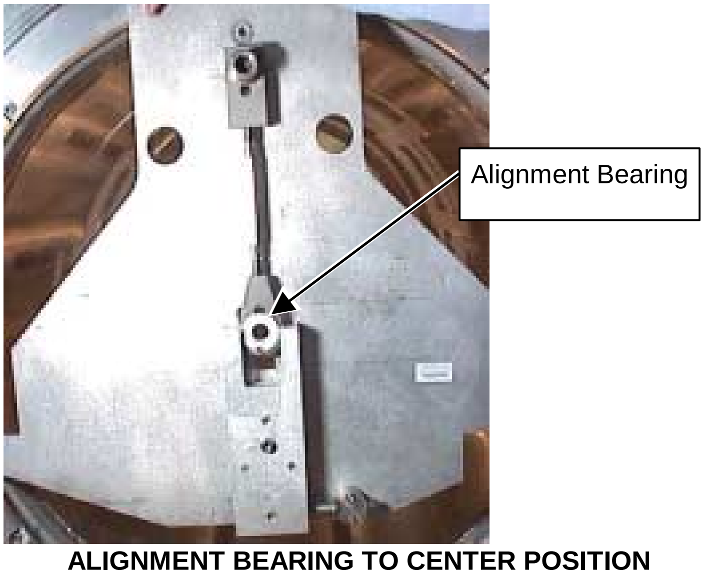
e. Place the Tube Support Plate Assembly on the Rear of the Magnet. See Illustration 3-9.

TUBE SUPPORT PLATE ASSEMBLY ON REAR OF MAGNET
ILLUSTRATION 3-9

f. Secure Plate Assembly by using a washer and nut on each stud and tightening till snug. See Illustration 3-10.

PLACING WASHER AND NUT ON STUD
ILLUSTRATION 3-10

g. Adjust the alignment bearing for center position as viewed from the side with the square cut out. See Illustration 3-11.



ALIGNMENT BEARING TO CENTER POSITION
ILLUSTRATION 3-11

6. Using Alcohol and a clean towel, wipe down the inside of the magnet. See Illustration 3-12.

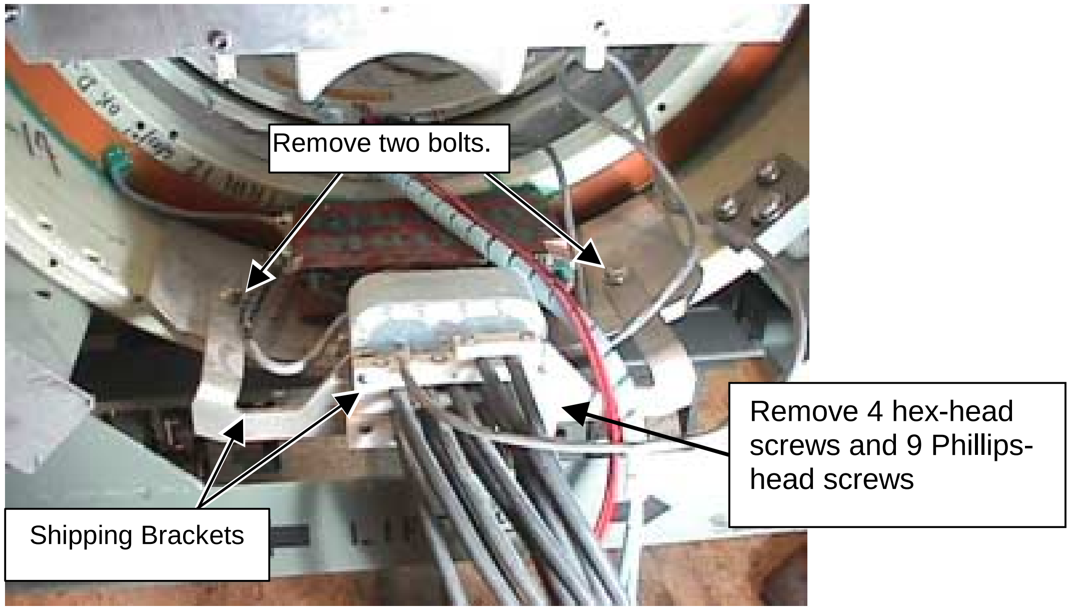
WIPING DOWN INSIDE OF MAGNET
ILLUSTRATION 3-12

7. If Gradient Support bracket is on the front and rear interface rings, remove bracket before installing the coil. The added clearance is needed for pulling the gradient cables through the bore.

WARNING!

POSSIBLE DAMAGE TO GRADIENT CABLE LEADS EXISTS. THE GRADIENT CABLE LEAD CONNECTIONS CAN BREAK EASILY. CAREFULLY REMOVE THE SHIPPING BRACKETS THAT SECURE THE GRADIENT CABLE LEADS.

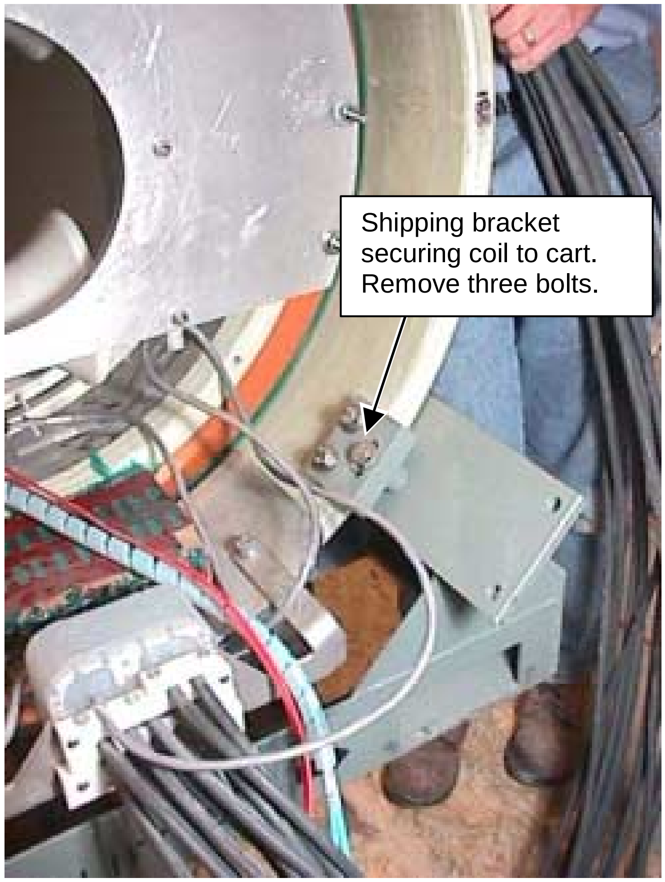
8. On the coil cart, remove the shipping brackets that hold the gradient cable leads in place. See Illustrations 3-13 and 3-14.



LOCATION OF SHIPPING BRACKETS HOLDING GRADIENT CABLES
ILLUSTRATION 3-13

SHIPPING BRACKETS REMOVED
ILLUSTRATION 3-14

9. Remove the shipping bracket that secures the coil to the cart. See Illustration 3-15.



LOCATION OF SHIPPING BRACKET
ILLUSTRATION 3-15

10. Place Male Tube Assembly in magnet and secure to the support plate using the cotter pin.
11. Place the Female Tube Assembly in the coil.
12. Maneuver cart with coil to the front of the magnet.

13. Carefully feed the gradient cables through the bore of the magnet and carefully pull out the back. See Illustration 3-16.

FEEDING GRADIENT COILS THROUGH MAGNET BORE
ILLUSTRATION 3-17

14. Remove shipping bolts from the cart. There are a total of 4 bolts, two on each side. See Illustration 3-17.

REMOVING SHIPPING BOLT FROM CART
ILLUSTRATION 3-17

15. Position the cart in front of the magnet and center the cart left/right in respect to the patient bore. Provide enough room for an FE to stand between the cart and the magnet so the FE can thread the support tube sections together.

16. Release the hand lever on the cart handle to set the brakes so the cart will not move after the cart is aligned to the magnet bore.

17. To properly align the coil, it may be necessary to move the cart up or down. See Illustration 3-18.

ADJUSTING HEIGHT OF CART
ILLUSTRATION 3-18

18. Connect the Tube Assembly. This takes one person at the rear of the magnet to hold the male tube; one person at the patient end of the coil and a person in the middle to make sure the tube connects.

19. Adjust the height of the cart so the end of the Tube is the same height as the Tube in the bore of the magnet.

20. Slowly thread the Tube sections together.

Note

Be very careful not to force the threads. Make vertical or horizontal adjustments, as necessary, to the alignment bearing or cart for an easy fit.

21. Remove the Gradient Coil Roller assemblies, quantity 2, from the tool crate. See Illustration 3-19.

GRADIENT COIL ROLLER ASSEMBLIES
ILLUSTRATION 3-19

22. Place roller assemblies on the coil, one on each side of the coil on the Patient end. See Illustration 3-20.

ROLLER ASSEMBLY ON COIL
ILLUSTRATION 3-20.

23. Go to the rear of the magnet. Adjust the Alignment Bearing in the vertical direction and watch for the tube to make contact to the upper roller on the rear Tube Guide Roller Assembly. Continue to raise the Alignment Bearing until the back end of the Gradient coil starts to raise off of the cradle.

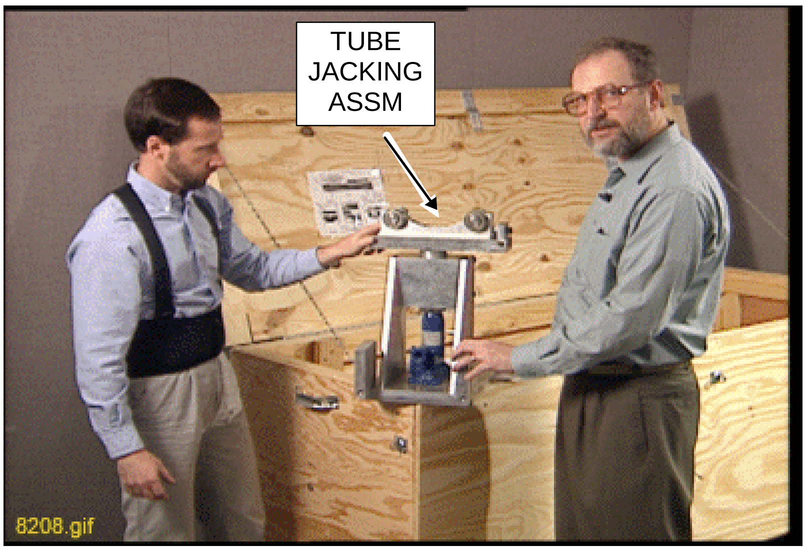
Note

When this happens the load at the back end of the Gradient coil is transferred from the cradle to the Support Tube and magnet.

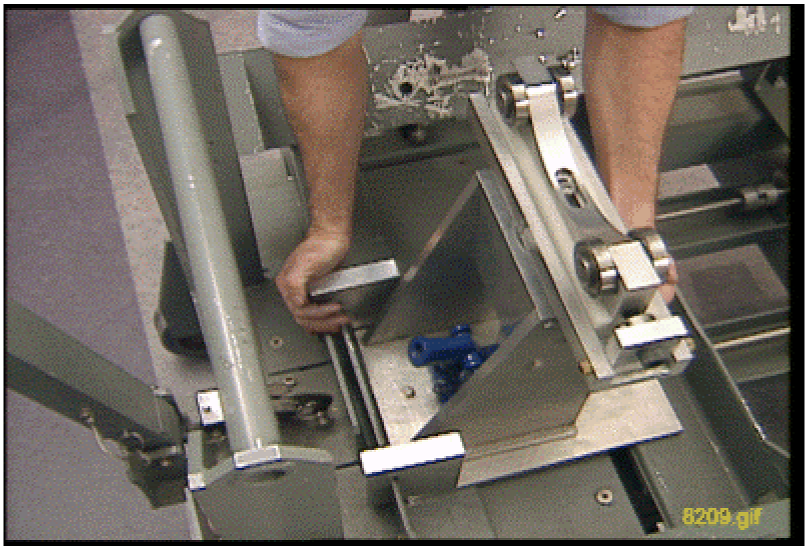
24. Move the rollers on the assemblies to the lowest position.

25. Slowly push the Gradient coil toward the magnet on the rollers until there is enough clearance to install the Tube Jacking assembly.

26. Remove the Tube Jacking Assembly from the crate. See Illustration 3-21. Install it onto cradle/cart. Secure with bolt through Tube Jack Assembly Base Plate. See Illustration 3-22.



TUBE JACKING ASSEMBLY
ILLUSTRATION 3-21



SECURE TUBE JACKING ASSEMBLY
ILLUSTRATION 3-22

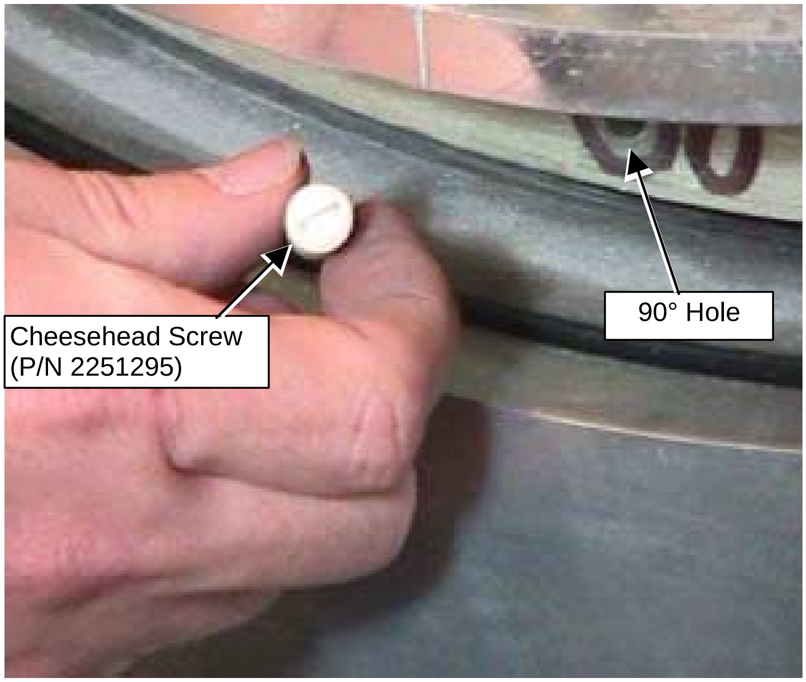
27. Operate the jack to raise the coil to put load on the upper rollers and allow for the removal of the Gradient coil Roller assemblies.

28. Remove the Gradient coil Roller Assemblies and return these parts to the shipping case.

29. Check for proper clearance around the TRM assembly in respect to the magnet bore opening. Make adjustments to the jack and alignment bearing to produce a fit that is symmetrical and level to the bore in front and back

30. Slowly push the Gradient coil into the magnet bore and watch the clearance at both ends. Carefully pull the gradient coils through the rear of the magnet. Continue to install until the Gradient coil is centered from front to back in the bore.

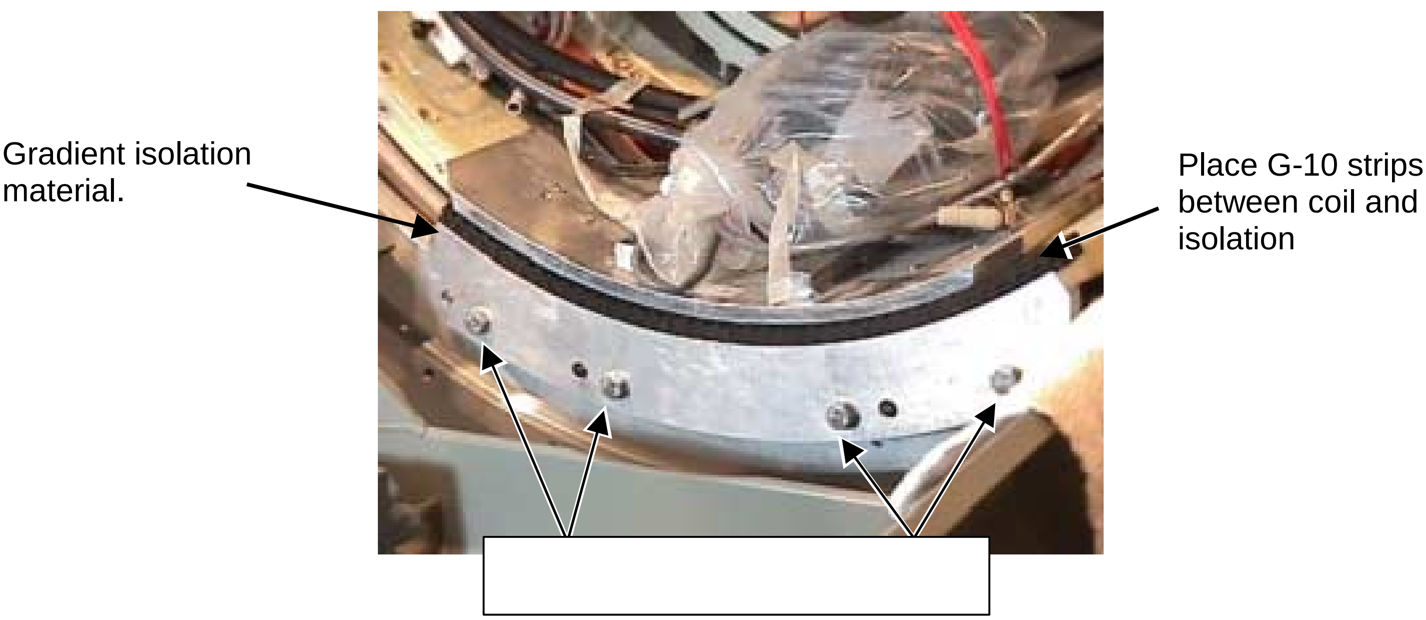
31. Install spacer by placing the cheesehead screw in the 90° hole. See Illustration 3-23.



INSTALLING SCREW INTO 90° HOLE
ILLUSTRATION 3-23

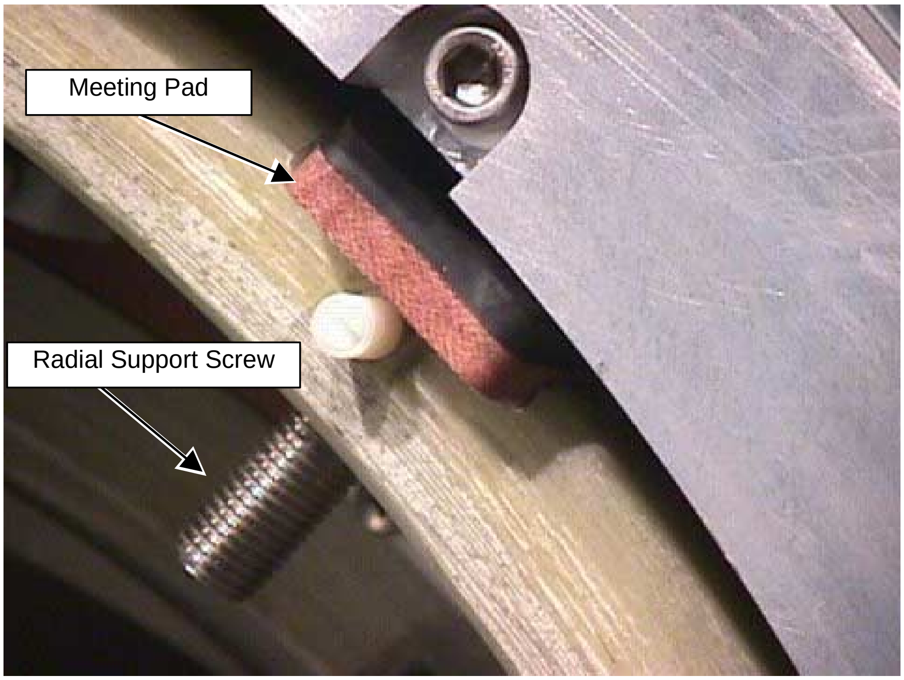
32. Install 15mm spacer on cheesehead screw so the screw head is flush or inside the spacer.

33. When the coil is centered in the Z direction, install the gradient isolation material (black rubber like stuff) between the coil and the support. You must use two interlocked strips per side. Use the supplied G-10 strips to center the coil in the Y direction. See Illustration 3-24



INSTALLING ISOLATION MATERIAL
ILLUSTRATION 3-24

34. Install the four Radial Supports between the TRM coil and the magnet warm bore. See Illustration 3-25. First tighten the Radial Support Screw until the Meeting Pad is secure to the magnet warm bore. Then tighten the Radial Support Screw an additional ½ turn. Then install the plastic set screw to prevent the Radial Support Screw from loosening.



INSTALLING RADIAL SUPPORTS
ILLUSTRATION 3-25

35. After installation and alignment is complete, disassemble the installation tool and return all parts to the crate.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	Oct. 2, 2001	K. Keshena	Preliminary version.
B	Oct. 27, 2001	K. Keshena	Included revisions from latest install.
0	April 6, 2002	P. Kargard	More feedback